HELP DESK TICKET RESOLUTION IMPROVEMENT

Functional Requirements Document (FRD)

1. Purpose and Functional Overview

This document defines the expected system behaviour for ticket lifecycle management, SLA monitoring, escalation notifications, and KPI reporting. It is intended to guide configuration, development, testing, and user acceptance. Field names and implementation details should be mapped to the selected ITSM platform.

2. Users and Permissions

Role	Expected access
Service desk agent	View permitted tickets; update assigned tickets; progress status; record resolution details.
Team lead	View team queue; monitor SLA risk; receive escalations; reassign tickets if authorised.
Service desk manager	View KPI dashboards and reports; approve operational rules.
ITSM administrator	Configure workflow, SLA rules, notifications, access, and dashboard settings.
Auditor / read-only user	View permitted ticket history and reports without modifying records.

3. Ticket Lifecycle and State Transitions

Status	Purpose / entry condition	Expected next step
New	Ticket has been created but not assessed.	Triage
Triage	Ticket is categorised, prioritised, and assigned.	In Progress / Resolution
In Progress	Agent or resolver team is actively working on the issue.	Resolved; optionally Waiting
Waiting for Requester / Third Party (optional)	Progress depends on external information or action.	In Progress when work resumes
Resolved	Solution or workaround is recorded and communicated.	Closed or Reopened if permitted
Closed	Closure conditions are met and the ticket is finalised.	Reopen only if policy permits
Reopened	Requester indicates the issue remains or has returned.	Triage / In Progress

Transition control: block invalid transitions where supported. For example, a ticket should not move from New directly to Closed without required triage and resolution information unless an approved exception exists.

4. Functional Requirements

ID	Function	Requirement	Acceptance check
FR-01	Ticket list	Display ticket ID, summary, status, priority, assignee/team, created/updated time, and SLA state where available.	Key fields are visible and help users identify tickets needing attention.

FR-02	Search and filters	Search by ticket ID/keywords and filter by status, priority, assignee/team, category, date, and SLA state.	Results match selected filters; clearing filters restores the default view.
FR-03	Ticket creation	Capture required fields such as summary, description, requester, category, and priority where applicable.	Valid submission creates a unique ticket in the configured initial status.
FR-04	Triage and assignment	Allow authorised users to categorise, prioritise, and assign tickets.	Changes save and appear in the queue and history.
FR-05	Status validation	Validate status changes against approved transitions and required fields.	Invalid transitions are blocked or show a clear validation message.
FR-06	SLA calculation	Calculate SLA time using approved targets, calendars, and pause/resume rules.	SLA status matches approved test cases.
FR-07	At-risk warning	Flag tickets that reach the configured pre-breach threshold.	Eligible tickets are flagged within the agreed processing interval.
FR-08	Escalation	Escalate at configured milestones to the agreed recipient or queue.	Rule, recipient, timestamp, and outcome are recorded.
FR-09	Notifications	Send configured notifications for escalation and approved workflow events.	Message includes ticket reference, summary, SLA state, and required action.
FR-10	Resolution details	Require resolution category and notes before Resolved when mandated by policy.	System blocks resolution if mandatory details are missing.
FR-11	Closure and reopening	Apply closure rules and allow reopening only under approved policy.	Status and timestamps update consistently.
FR-12	Audit history	Record status, assignee, priority, escalation, resolution, and closure changes.	Authorised users can view chronological history.
FR-13	KPI dashboard	Display ticket volume, MTTR, SLA compliance, open-ticket breaches, and trends.	Metrics follow approved definitions and reconcile with source data.
FR-14	Dashboard filters	Support date, priority, team, category, and status filters where available.	Metrics update consistently when filters change.
FR-15	Data refresh	Refresh metrics on an agreed schedule or on demand; show last refresh where practical.	Stale data is identifiable and refresh failures are logged or surfaced.
FR-16	Access control	Restrict ticket updates, configuration, and reports according to role.	Users can perform only permitted actions.
FR-17	Export (if approved)	Allow authorised export/sharing using platform-supported controls.	Exports respect filters and access permissions.

5. SLA and Escalation Logic

The service owner must approve all thresholds. The following table is a proposed rule pattern, not a replacement for contractual SLA commitments.

Condition	System action	Recipient	Expected record

Warning threshold reached	Mark At Risk and show remaining SLA time.	Assigned agent / resolver queue	Warning timestamp and rule
Escalation threshold reached	Send notification and flag ticket in team lead view.	Team lead / designated queue	Escalation timestamp, recipient, and reason
SLA target exceeded	Mark Breached and notify the approved escalation recipient.	Team lead / service owner per policy	Breach timestamp and SLA target
Approved waiting state	Pause or adjust SLA clock only if policy allows.	No alert unless configured	Pause reason, start time, and resume time

6. KPI Dashboard Specifications

KPI	Proposed calculation	Display / drill-down
Ticket volume	Count created, resolved, reopened, and currently open in the selected period.	Summary cards and time trend; filter by priority/team.
MTTR	Average time between agreed start event and resolution timestamp for eligible resolved tickets.	Trend and breakdown by priority, category, or team.
SLA compliance	Eligible tickets resolved within SLA ÷ all eligible resolved tickets × 100.	Percentage and trend; show numerator/denominator where practical.
Open-ticket SLA breaches	Count of currently open tickets that exceeded their applicable SLA target.	Count and list of age, priority, assignee, and ticket reference.
At-risk tickets	Open tickets that crossed warning threshold but have not breached.	Queue sorted by remaining SLA time or priority.

Metric definitions require approval, including business versus calendar hours, paused time, reopened tickets, cancelled/duplicate tickets, reporting time zone, and the handling of tickets resolved outside the selected period.

7. Wireframe Guidance

A. Service Desk Queue

- Top counters: Open, At Risk, Breached, Unassigned, and Resolved Today.
- Main table: Ticket ID, summary, priority, status, assignee, age, SLA remaining/breached, and last updated.
- Filters: date range, status, priority, assignee/team, category, and SLA state.
- Use clear warning and breach indicators; never rely on colour alone.

B. Service Performance Dashboard

- KPI cards for ticket volume, MTTR, SLA compliance, open breaches, and at-risk tickets.
- Trend charts for created versus resolved tickets and SLA compliance over time.
- Breakdowns by priority, category, and resolver team to highlight bottlenecks.
- Drill-down from a metric to the underlying tickets, subject to permissions.

8. Data Requirements

Field	Purpose	Validation / consideration
Ticket ID	Unique reference for search, audit, and drill-down.	Unique and persistent.
Created / updated timestamps	Ticket age, volume trends, and workflow history.	Consistent timezone and reporting cut-off.
Status and status history	Lifecycle visibility and transition validation.	Map values to approved lifecycle.

Priority and category	Queue ordering and reporting breakdowns.	Agree allowed values and mapping.
Assignee / resolver team	Ownership and escalation routing.	Unassigned tickets must be identifiable.
SLA target and timestamps	Remaining time, breach state, and compliance.	Define calendar, pause, and reset rules.
Resolved / closed timestamps	MTTR and closure reporting.	Use agreed start and stop events.
Resolution code / notes	Resolution quality and closure controls.	Required fields defined by policy.

9. Non-Functional Requirements

- Performance: lists and dashboards should meet response-time targets agreed with the ITSM owner.
- Availability: follow the service desk platform's approved availability and support arrangements.
- Security: enforce role-based access and existing data-handling policies.
- Usability: use clear labels, statuses, and actionable warning messages.
- Auditability: material workflow changes and escalation events must be traceable.
- Maintainability: authorised administrators should be able to configure thresholds and recipients where supported.
- Accessibility: communicate status using text/icons as well as colour.

10. End-to-End Acceptance Scenarios

AT-01 — New ticket: A valid submission creates a unique ticket in the configured initial status and records creation time.

AT-02 — Triage and assignment: An authorised agent updates category, priority, and assignee; the queue and history reflect the changes.

AT-03 — At-risk alert: When an open ticket reaches the warning threshold, it is flagged At Risk and the configured notification is sent if enabled.

AT-04 — SLA breach: When an eligible ticket exceeds its target, it is marked Breached and the escalation event is recorded.

AT-05 — Invalid transition: If mandatory resolution data is missing, the system blocks resolution or identifies the missing fields.

AT-06 — Closure: When closure conditions are met, closing the ticket records the status and closure timestamp.

AT-07 — Dashboard reconciliation: Dashboard counts and SLA metrics reconcile with source tickets using approved definitions.

AT-08 — Role permissions: A user without configuration permission cannot change SLA or workflow settings.

11. Open Questions for Configuration

- Which ITSM platform and edition will be used?
- What are the approved SLA targets by priority/service/type?
- Which business calendar, holidays, and pause conditions apply?
- What warning and escalation thresholds and recipients are required?
- Can resolved tickets be reopened, and under which conditions?
- How are MTTR, SLA compliance, and open-ticket breach rate defined?
- What baseline and post-implementation periods support the reported 25% reduction?

- How frequently should data refresh, and is export required?

12. Suggested Test Evidence

- Workflow and state transition test results.
- SLA warning, escalation, pause/resume, and breach test results.
- Notification delivery and recipient validation.
- Role and permission test results.
- Dashboard reconciliation against sampled source tickets.
- User acceptance sign-off from service desk agents and manager.

Document status: Draft for functional review. Confirm platform capabilities, SLA rules, and KPI definitions before build.